

Predicting Calories Burned During Exercise: A Regression Pipeline with Model Comparison, Robustness, and Interpretability Analysis

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Abstract—Fitness trackers commonly estimate calorie expenditure using generic heuristics that ignore individual physiology. This paper presents a supervised regression pipeline that predicts calories burned during a workout from biometric and exercise-session data (15,000 records). Outlier bounds and feature scaling are fit exclusively on a training partition and applied unchanged to a held-out test partition. Six regression models are compared, the three strongest are tuned via cross-validated search, and the resulting gains are confirmed with a repeated-split robustness check. The final tuned gradient boosting model reaches $R^2 = 0.9995$ and a mean absolute error of 0.96 kcal on the test set. Feature importance and SHAP analysis show that a single engineered feature, workout workload, dominates the prediction. Because all candidates are compared on the same test partition rather than a separate validation set, the reported score is discussed as an optimistic estimate.

Index Terms—calorie prediction, regression, feature engineering, gradient boosting, hyperparameter tuning, SHAP, machine learning.

1 INTRODUCTION

WEARABLE fitness devices and gym equipment routinely report an estimate of calories burned during a workout. In practice, many such estimates rely on coarse, population-level heuristics that map exercise type and duration to an average calorie cost, largely ignoring how an individual's heart rate, body temperature, age, or body composition affect true energy expenditure. A data-driven regression model trained on paired session and outcome data offers the possibility of a substantially more personalized estimate.

This paper addresses the task of predicting Calories, a continuous target measured in kilocalories, from a combination of static attributes (age, height, weight, gender) and session-level biometric signals (exercise duration, heart rate, body temperature). The data come from two files, `exercise.csv` and `calories.csv` [4], joined on a user identifier, yielding 15,000 complete records with no missing values, duplicates, or physiologically implausible readings.

Beyond fitting an accurate model, this work aims for transparency about how data leakage was avoided. All data preparation steps, namely outlier bounds and feature scaling, are fit exclusively on the training partition and applied unchanged to the test partition, so that no information about test rows influences those choices. Model selection is the weaker point: every baseline, tuned, and ensemble candidate is compared on the same held-out test partition rather than on a separate validation set, which can optimistically bias the reported score of the winning candidate. This limitation is stated explicitly, and a corrected protocol is proposed as future work (Section IV).

The main contributions of this paper are:

- A leakage-safe data preparation pipeline, in which

outlier bounds and scaling are fit exclusively on training data before being applied to the test partition.

- An engineered workload feature (duration multiplied by heart rate) that substantially improves on the raw features taken individually.
- A systematic comparison of six regression models, cross-validated hyperparameter tuning for the three strongest candidates, and a repeated-split robustness check that shows the tuning gains are consistent across splits rather than an artifact of a single split.
- An interpretability analysis combining impurity-based feature importance with SHAP values.
- A discussion of test-set reuse during model selection as a limitation of the evaluation protocol, together with a concrete proposal for correcting it.

1.1 Group Organization

This project was completed individually. The author was solely responsible for every stage of the work reported here: problem definition, data understanding and quality assessment, feature engineering, exploratory data analysis, model implementation, hyperparameter tuning, the robustness and interpretability analyses, and the writing of this paper.

1.2 Paper Organization

The remainder of this paper is organized as follows. Section II describes the proposed method, including data preparation, feature engineering, model selection, and evaluation protocol. Section III reports the resulting metrics and the main insights drawn from them. Section IV concludes and discusses limitations and future work.

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2 PROPOSED METHOD

2.1 Data Understanding and Quality

The merged dataset contains nine columns per session: the identifier `User_ID`, seven attributes (`Gender`, `Age`, `Height`, `Weight`, `Duration`, `Heart_Rate`, `Body_Temp`), and the target `Calories`, which ranges from 1 to 314 kcal. `Gender` is nearly balanced (50.4% female, 49.6% male). Five formal data-quality checks were run: completeness, representation of expected categorical values, plausibility of numeric ranges (e.g., age between 10 and 100, heart rate between 30 and 220 bpm), row/identifier uniqueness, and unexpected zero values. All five checks passed cleanly: zero missing values across all 15,000 rows, zero duplicate rows and zero duplicate `User_ID` values, zero rows failing any plausibility check, and zero unexpected zero-values in any numeric column. This indicates the dataset is well curated; the checks therefore serve mainly as a template that would catch real issues on noisier, real-world wearable exports.

2.2 Feature Engineering

Two derived features were added: body mass index, $BMI = \text{Weight}/(\text{Height}/100)^2$, and a workload proxy, $\text{Workload} = \text{Duration} \times \text{Heart_Rate}$, intended to capture overall cardiovascular effort rather than duration or intensity alone. `Gender` was encoded as a binary `Gender_Encoded` feature (male = 1, female = 0); with only two categories, a single 0/1 column carries the same information as one-hot encoding without a redundant column. `User_ID` carries no predictive meaning and was dropped. Pearson correlation with the target confirmed that `Workload` ($r = 0.975$), `Duration` (0.955), `Heart_Rate` (0.898), and `Body_Temp` (0.825) carry almost all of the linear signal, while static attributes such as `Age`, `BMI`, `Weight`, `Height`, and `Gender_Encoded` are only weakly correlated ($|r| < 0.16$).

2.3 Exploratory Data Analysis

The target `Calories` is moderately right-skewed (skewness 0.51), with a long tail toward high-effort sessions. No log transformation of the target was applied, since the skew is moderate and the best-performing models are tree-based and insensitive to the shape of the target distribution. Boxplots show different outlier patterns: `Heart_Rate` has a single upper-tail point (128 bpm), `Body_Temp` a cluster of lower-tail values (short sessions with lower body temperature), and `BMI` a few outliers in both tails, while `Duration` and `Workload` show none. This motivates the outlier-clipping step described next. Scatterplots of `Duration`, `Heart_Rate`, and `Workload` against `Calories` each show a strong, roughly linear relationship, consistent with the correlation coefficients reported above, and a full correlation heatmap reproduces the same ranking.

2.4 Data Preparation

The data were split into training and test subsets (80/20) before any outlier handling or scaling, so that no information about test rows could influence decisions made on the training set. Outlier bounds for `Heart_Rate`, `Body_Temp`, and `BMI`, the features with outliers in the boxplots, were

computed with the interquartile-range (IQR) rule using training rows only, then applied unchanged to clip the test rows. The IQR rule was preferred over the three-sigma rule because it does not assume normally distributed data and quartiles are robust to the extreme values themselves. Values were clipped (winsorized) rather than deleted: all of them passed the plausibility checks, so they are rare but real measurements, and clipping limits their influence without losing rows. Feature scaling (standardization) was likewise fit only on the training set with a `scikit-learn` [1] `ColumnTransformer`, then applied to the test set via `transform` alone. Scaling is required by k -NN, which relies on distances, and helps the linear models. A single shuffled 5-fold `KFold` split of the training set is shared by every cross-validation step, so all models are compared on identical folds.

2.4.1 Model-Selection Protocol

No separate validation set was carved out of the training data. Baseline models, the polynomial-feature experiment, hyperparameter tuning (selected via 5-fold cross-validation within the training set), and the voting ensemble are all subsequently evaluated on the same held-out test partition, and the best-performing configuration is chosen by comparing those test metrics directly. The test set is therefore queried for roughly a dozen candidate configurations, which can optimistically bias the reported metric of the winner. This trade-off is discussed further in Section IV.

2.5 Feature Selection

Given the modest dimensionality (nine candidate features), aggressive reduction (e.g., PCA) was not pursued. Instead, Recursive Feature Elimination (RFE) with a linear regression estimator was swept over subset sizes from two to nine features, scored via 5-fold cross-validation on the training set. Cross-validated R^2 was 0.962, 0.963, 0.966, and 0.967 at two, three, four, and five features respectively, then jumped to 0.984 at six features and stayed within 0.0001 of that value through nine, indicating no feature was redundant enough to discard. All nine features were therefore retained for every downstream model; RFE served here as a diagnostic rather than a filtering step.

2.6 Models and Training

Six regression models were trained as baselines on the training set and evaluated on the test set: Decision Tree (max depth 6), k -Nearest Neighbors ($k = 15$), Linear Regression, Ridge regression ($\alpha = 1$), Random Forest (300 trees, max depth 8), and Gradient Boosting [3] with default settings (100 trees, depth 3, learning rate 0.1). A degree-2 polynomial feature expansion was additionally tested with Linear Regression to probe whether the true `Duration/Heart-Rate-Calories` relationship has curvature that a plain hyperplane misses.

The three strongest model families, Random Forest, Gradient Boosting, and k -NN, were then tuned with 5-fold cross-validation on the training set, scored by R^2 . Randomized search sampled 5 of the 6 forest configurations (tree count and depth) and 5 of the 8 boosting configurations (tree count, learning rate, and depth), while an exhaustive grid

TABLE 1
Baseline Model Comparison (Test Set)

Model	R^2	RMSE (kcal)	MAE (kcal)
Gradient Boosting	0.9979	2.91	2.13
Random Forest	0.9953	4.36	3.04
k -NN	0.9946	4.65	3.33
Linear Regression	0.9845	7.90	5.83
Ridge	0.9845	7.91	5.81
Decision Tree	0.9818	8.58	6.06

search covered all 10 k -NN configurations (neighborhood size and uniform or distance weighting), which are cheap to train. The best configuration of each search was refit on the full training set.

2.7 Robustness and Interpretability

To verify that the improvement from tuning was not an artifact of the particular train/test split used for the search, each tuned configuration was refit and re-evaluated against its untuned baseline across ten different random splits (with outlier clipping and scaling refit fresh on every split). The mean R^2 difference was reported with a 95% confidence interval (mean $\pm$ 1.96 standard errors). This check measures whether the tuning effect itself is consistent across splits; it does not address the model-selection caveat above.

Interpretability was assessed two ways: impurity-based feature importance for the tuned tree ensembles, and SHAP (SHapley Additive exPlanations) [2] values computed with a tree explainer on a 200-row test-set sample, cross-checked against the impurity ranking, since impurity-based importance can be biased when features are correlated. A learning curve of the best model and a voting ensemble of the three tuned models were also evaluated.

3 RESULTS

3.1 Baseline and Tuned Model Comparison

Table 1 reports test-set performance for the six baseline models. Gradient Boosting led from the outset ($R^2 = 0.9979$), followed by Random Forest and k -NN, with the linear models clearly behind the non-linear ones. Five-fold cross-validation on the training set reproduced the same ranking ($R^2 = 0.9974 \pm 0.0001$ for Gradient Boosting), confirming it was not a single-split artifact. The degree-2 polynomial expansion improved Linear Regression substantially (R^2 : 0.9845 $\rightarrow$ 0.9987; MAE: 5.83 $\rightarrow$ 1.72 kcal), showing that the relationship has real curvature that a purely linear model misses.

Table 2 shows the effect of hyperparameter tuning. Gradient Boosting again led, reaching $R^2 = 0.9995$ (MAE = 0.96 kcal) with 400 trees, depth 5, and a learning rate of 0.1. For both ensembles the best values lie at the edge of the searched grid, so better configurations may exist beyond it. The repeated-split robustness check showed that all three tuning gains are consistent across splits, with 95% confidence intervals excluding zero: Random Forest +0.0023 R^2 (CI [0.0022, 0.0024]), Gradient Boosting +0.0021 (CI [0.0020, 0.0022]), and k -NN +0.0004 (CI [0.0004, 0.0004]).

A voting ensemble of the three tuned models reached $R^2 = 0.9988$ (MAE = 1.51 kcal), worse than the tuned

TABLE 2
Effect of Hyperparameter Tuning (Test Set)

Model	Base R^2	Tuned R^2	Params
k -NN	0.9946	0.9950	$k=15$, dist.
Random Forest	0.9953	0.9974	250 trees, $d=10$
Gradient Boosting	0.9979	0.9995	400 est., $d=5$

TABLE 3
Feature Importance, Tuned Models (Impurity-Based)

Feature	Random Forest	Gradient Boosting
Workload	0.9578	0.9414
Age	0.0254	0.0259
Gender_Encoded	0.0075	0.0083
Heart_Rate	0.0065	0.0209
Weight	0.0017	0.0032
Height	0.0006	0.0000
BMI	0.0003	0.0000
Body_Temp	0.0001	0.0000
Duration	0.0001	0.0001

Gradient Boosting model alone, since equal-weight averaging pulls the best model toward the comparatively weaker k -NN and Random Forest predictions. The learning curve of the tuned Gradient Boosting model shows training R^2 close to 1.0, while validation R^2 rises from about 0.9956 with roughly 1,000 examples to about 0.9995 with the full training fold; the gap narrows steadily, with no sign of overfitting.

3.2 Final Model Selection

Comparing all baseline, tuned, and ensemble candidates on the same held-out test set, tuned Gradient Boosting is the best-performing configuration: $R^2 = 0.9995$, RMSE = 1.39 kcal, MAE = 0.96 kcal, i.e., an average error below 1 kcal on a 1–314 kcal range, which clearly meets the success criteria. As noted in Section II, this number results from comparing roughly a dozen candidates on the same test partition, so it should be read as a slightly optimistic estimate of the model’s generalization performance.

3.3 Feature Importance and SHAP

Impurity-based importance and mean absolute SHAP values agree closely for both tuned tree ensembles. Table 3 reports the full impurity-based breakdown. *Workload* overwhelmingly dominates both models (mean $|\text{SHAP}| \approx 50$ –54 kcal), with *Age* and *Heart_Rate* far behind. Most features’ SHAP-versus-impurity ranks differ by at most one position; the one exception is *Height* in Gradient Boosting, which impurity ranks 7th but SHAP ranks 9th (last), a two-position shift. Overall, the collinearity between *Workload*, *Duration*, and *Heart_Rate* does not meaningfully distort the impurity-based ranking. The practical implication is that most of the model’s predictive power is concentrated in a single engineered interaction term rather than spread across many raw measurements, consistent with the near-linear scatter of *Workload* against *Calories* observed during exploratory analysis.

4 CONCLUSIONS

This paper presented a regression pipeline for predicting exercise calorie expenditure, reaching $R^2 = 0.9995$ (MAE

= 0.96 kcal) with a tuned Gradient Boosting model on the test set. Data preparation is leakage-safe: outlier bounds and scaling are fit exclusively on the training partition. Feature engineering played an outsized role: a single derived feature, workload, was found by both impurity-based importance and SHAP to dominate the predictions, and the repeated-split robustness check showed that the hyperparameter-tuning gains for all three tuned models are consistent across splits.

The most important limitation of this study is methodological: model selection compares every baseline, tuned, and ensemble candidate on the same held-out test set rather than a separate validation partition, so the reported $R^2 = 0.9995$ is best read as an optimistic estimate. A concrete fix for future work is a third, validation-only partition (e.g., a 60/20/20 train/validation/test split), on which all comparisons are run, touching the test set only once for the chosen model. Because Gradient Boosting was the best model at every stage of this study, including cross-validation and the repeated splits, the ranking of the models is likely to hold under such a protocol, even if the reported R^2 decreases slightly.

Two further limitations are worth noting. First, the dataset is unusually clean (no missing values, duplicates, or implausible readings), so results may not transfer directly to noisier, real-world wearable-device data. Second, Heart_Rate and Body_Temp are physiological responses that are only observable during or after a workout; the model is consequently better understood as estimating calories for a completed session rather than forecasting them before a workout begins. A natural extension is an ex-ante variant of this pipeline restricted to attributes known in advance (age, height, weight, gender, planned duration), which would likely be a harder and more product-relevant prediction task. Finally, the best hyperparameters lie at the edge of the searched grids, so wider ranges or Bayesian optimization could be explored.

REFERENCES

- [1] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems* 30, 2017, pp. 4765–4774.
- [3] J. H. Friedman, “Greedy Function Approximation: A Gradient Boosting Machine,” *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
- [4] Kaggle, “Calories Burnt Prediction Dataset (exercise.csv, calories.csv),” <https://www.kaggle.com/>, accessed 2026.